

Sri Lanka Institute of Information Technology

Data warehousing and Business

Intelligence

Assignment 2



KUMARAGE C D – IT23572560

Year 03 Semester 01 - WE 01.02

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Step 1: Data Source

1.1 Overview of the Data Source

For Assignment 2, the data source used is the Retail Data Warehouse (RetailDW) that was designed, implemented, and populated with data in Assignment 1. The data warehouse was built using SQL Server Integration Services (SSIS) and Microsoft SQL Server Management Studio (SSMS) with SQL Server as the target database platform.

1.2 Data Warehouse Details

Database Name	RetailDW
Platform	SQL Server
ETL Tool	SQL Server Integration Services (SSIS)
Development Tool	SQL Server Management Studio (SSMS)
Data Sources	CSV files (customers, reviews, support tickets), Excel files (campaigns, interactions), SQL Server table (transactions)
Schema Type	Star Schema
Data Range	2020 – 2025

1.3 Tables Used

1.3.1 Fact Table

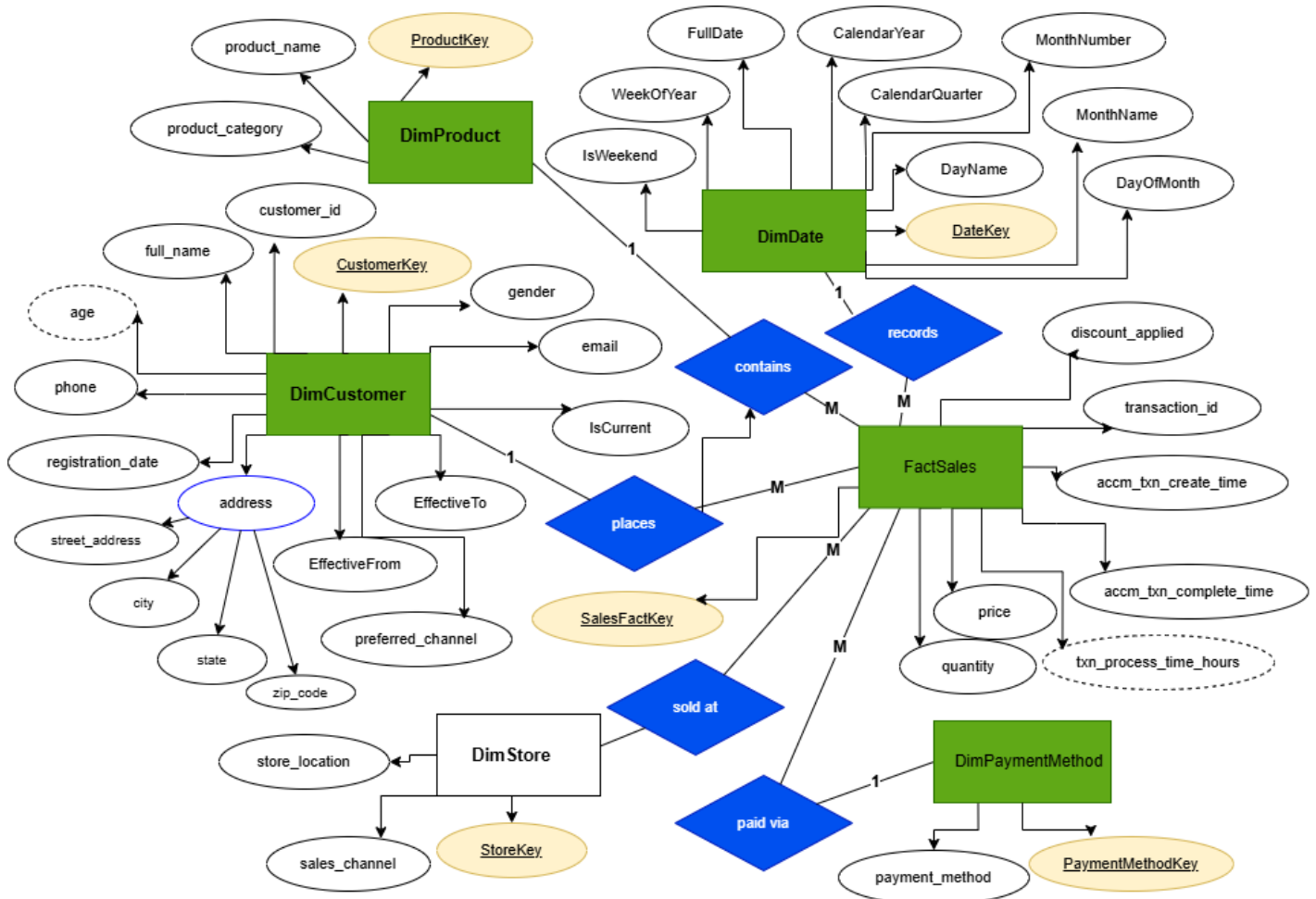
- FactSales – contain transactional sales data including measures such as quantity, price, discount_applied, and txn_process_time_hours

1.3.2 Dimension Tables

- DimDate – contain date-related attributes for time analysis.
- DimCustomer – contain customer demographic and geographic details
- DimProduct – contain product and product category information
- DimStore – contains store location and sales channel information.
- DimPaymentMethod – contain payments method classifications

1.4 ER Diagram

The following ER diagram represent the RetailDW data warehouse. It illustrates the relationships between the central fact table and the surrounding dimension tables. how the relationships works in RetailDW.



[\(view ER diagram\)](#)

1.5 Data Preparation Verification

Before using the data warehouse as source for Assignment 2, the following verifications and fixes were applied to ensure data integrity:

- Verified all dimension and fact tables are properly loaded
- Confirmed all foreign key relationships are intact
- Ensured no NULL keys exist in the fact table

Step 2: SSAS Cube Implementation

2.1 Creating the SSAS Project

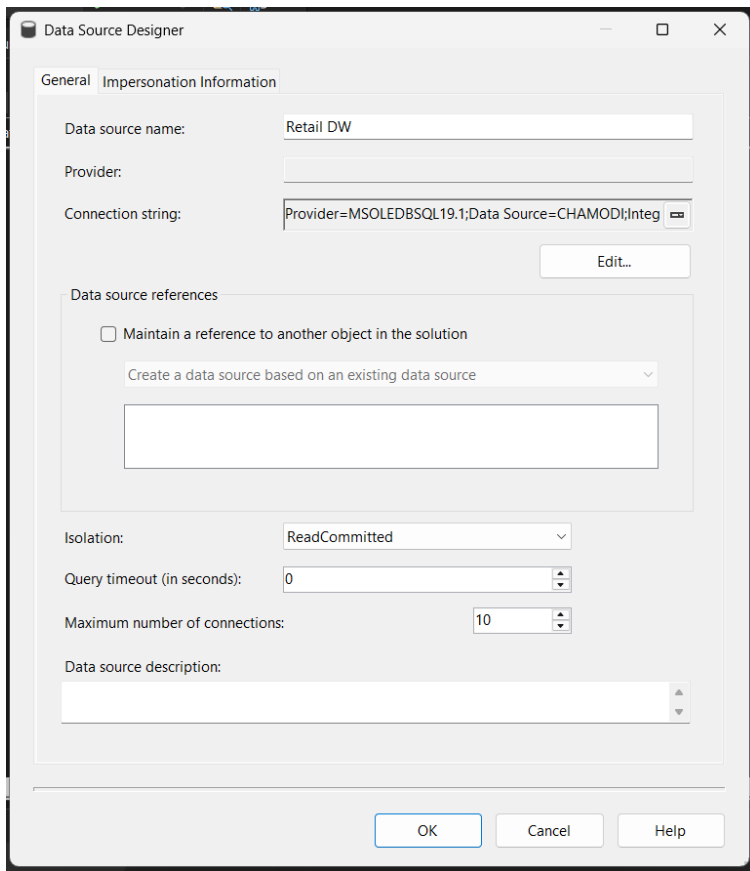
SSAS Multidimensional Cube Project is created using SQL Server Data Tools within Visual Studio. The following steps are followed:

1. Open Visual Studio and create new Analysis Services Multidimensional Project
2. Set the project name as RetailCubeProject and Click Create to initialize the project.

2.2 Creating the Data Source

Data source connection was established to link the SSAS project to the RetailDW data warehouse:

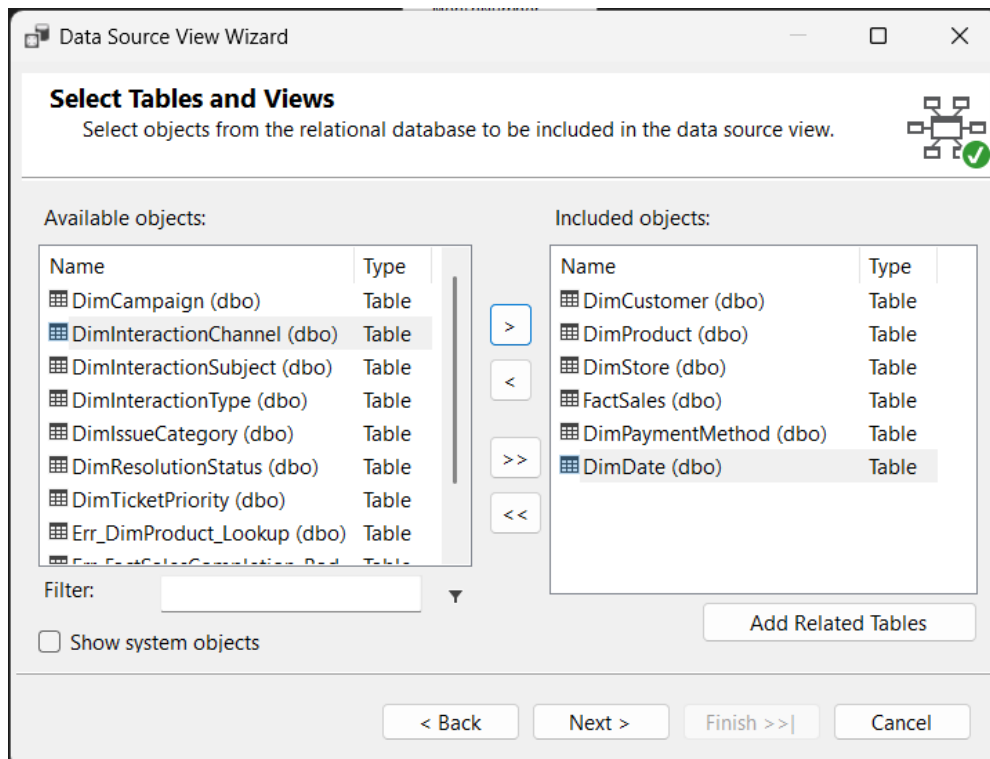
1. Create New Data Source to launch the wizard and Select SQL Server as the connection type
2. Set Server to: CHAMODI and Set Database to: RetailDW
3. Select Windows Authentication (chamodi\msi) as the authentication method and Complete the wizard and save the data source



2.3 Creating the Data Source View (DSV)

Data Source View was created to define the logical schema used by the cube:

1. Create New Data Source View and Select the RetailDW data source
2. Add the following tables to the DSV:
 - FactSales
 - DimDate
 - DimCustomer
 - DimProduct
 - DimStore
 - DimPaymentMethod
5. Complete the wizard to generate the DSV



2.4 Creating Dimensions

Individual dimensions were created for each dimension table in the DSV. The following dimensions were configured:

2.4.1 DimDate Dimension

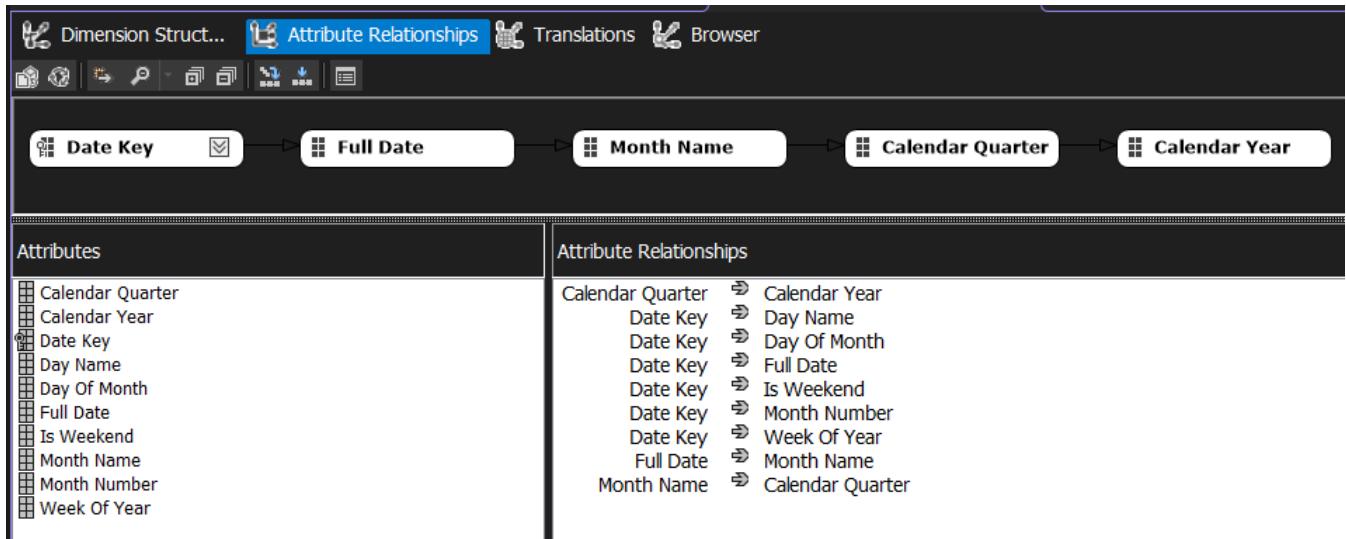
Attributes used:

- DateKey (Key Attribute)
- FullDate
- CalendarYear
- CalendarQuarter
- MonthNumber
- MonthName

2.4.1.1 Date Hierarchy

Hierarchy Name: Calendar Hierarchy

- Level 1: Calendar Year
- Level 2: Calendar Quarter
- Level 3: Month Name
- Level 4: Full Date



2.4.2 DimProduct Dimension

Attributes used:

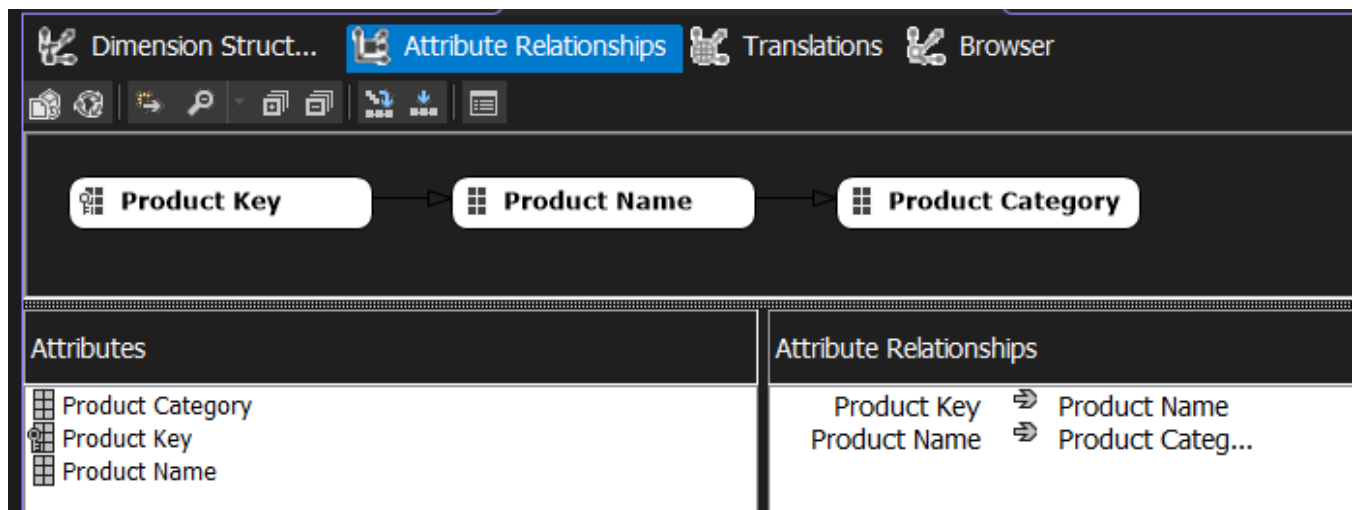
- ProductKey (Key Attribute)
- product_name
- product_category

Attribute	KeyColumns	NameColumn
Product Name	ProductKey	product_name
Product Category	product_category	product_category

2.4.2.1 Product Hierarchy

Hierarchy Name: Product Hierarchy

- Level 1: Product Category
- Level 2: Product Name



2.4.3 DimCustomer Dimension

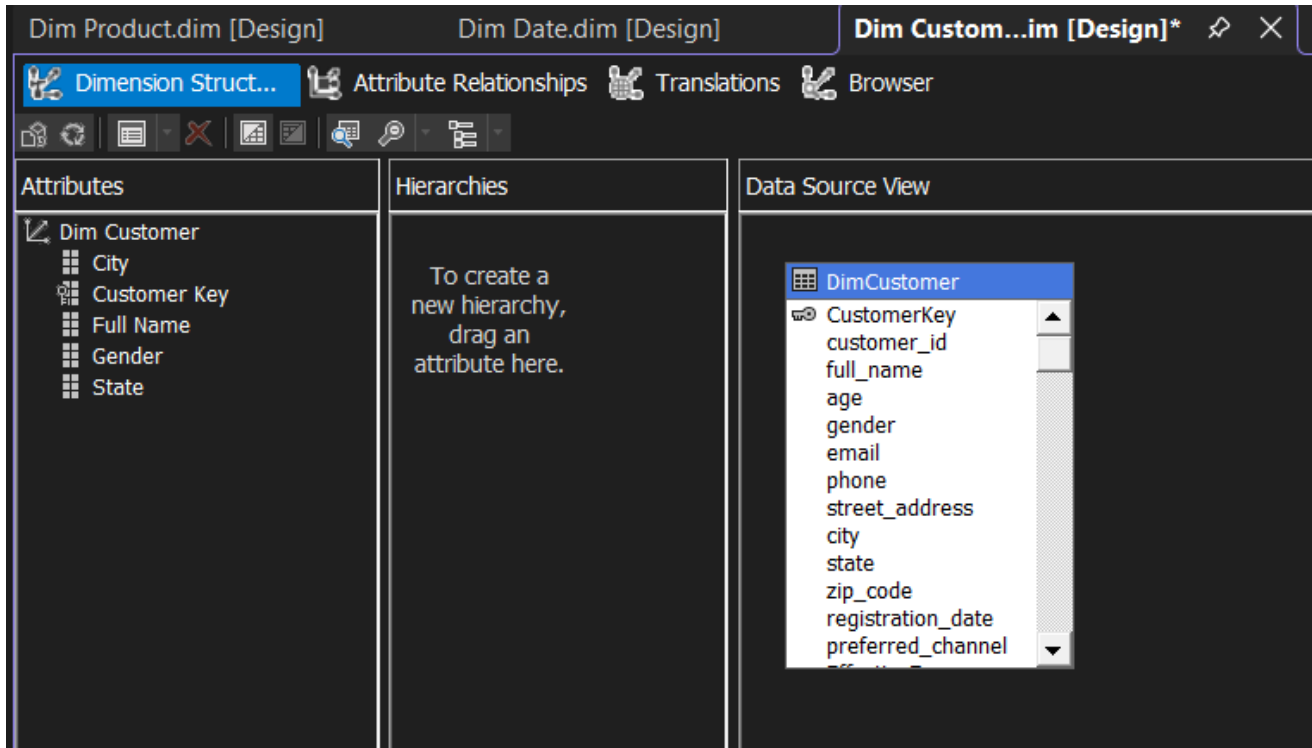
No hierarchies are created for DimCustomer.

Attributes used:

- CustomerKey (Key Attribute)
- full_name
- city
- state

Attribute	KeyColumns	NameColumn
Full Name	CustomerKey	full_name
City	CustomerKey	city
State	CustomerKey	state

Attributes Relationships used in DimCustomer :

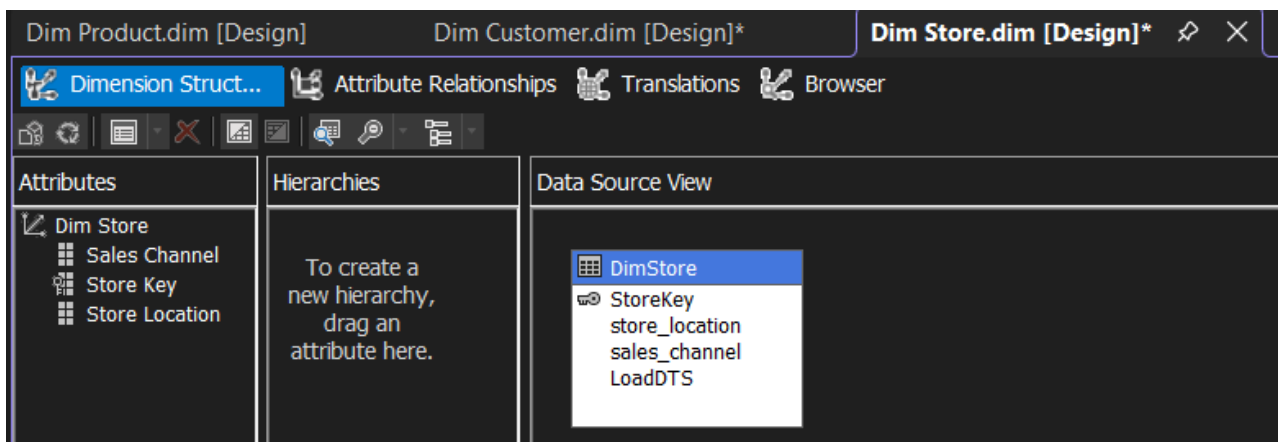


2.4.4 DimStore Dimension

No hierarchies are created for DimStore and DimPaymentMethod as these dimensions do not have a natural hierarchical structure

Attributes used:

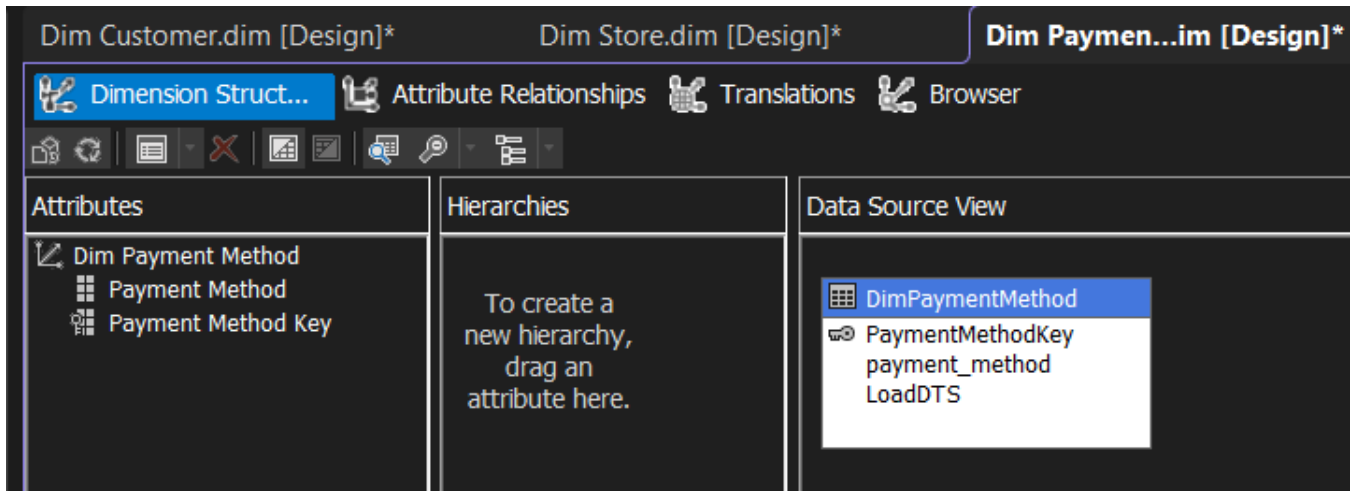
- StoreKey (Key Attribute)
- store_location
- sales_channel



2.4.5 DimPaymentMethod Dimension

Attributes used:

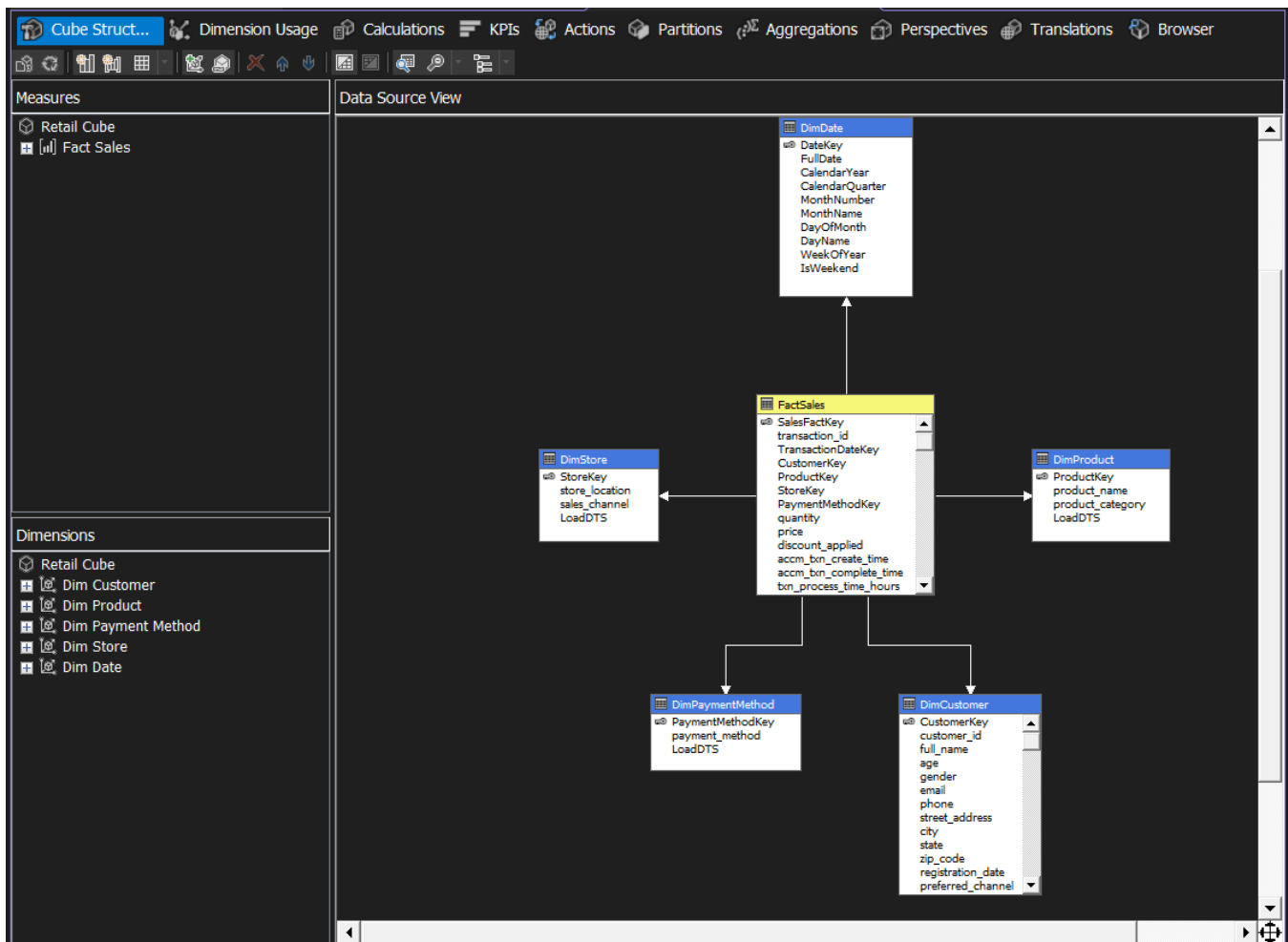
- PaymentMethodKey (Key Attribute)
- payment_method



2.5 Creating the Cube

The cube was created using the Cube Wizard in SSDT

- 1.Right-click on Cubes in Solution Explorer Click New Cube
- 2.Select Use existing tables as the build method
- 3.Select FactSales as the measure group table
- 4.Add the following dimensions to the cube:
 - Date (DimDate)
 - Customer (DimCustomer)
 - Product (DimProduct)
 - Store (DimStore)
 - Payment Method (DimPaymentMethod)
- 5.Complete the wizard



2.6 Configuring Dimension Usage

Dimension relationships were configured in the Dimension Usage tab to link each dimension to the fact table via the appropriate foreign key

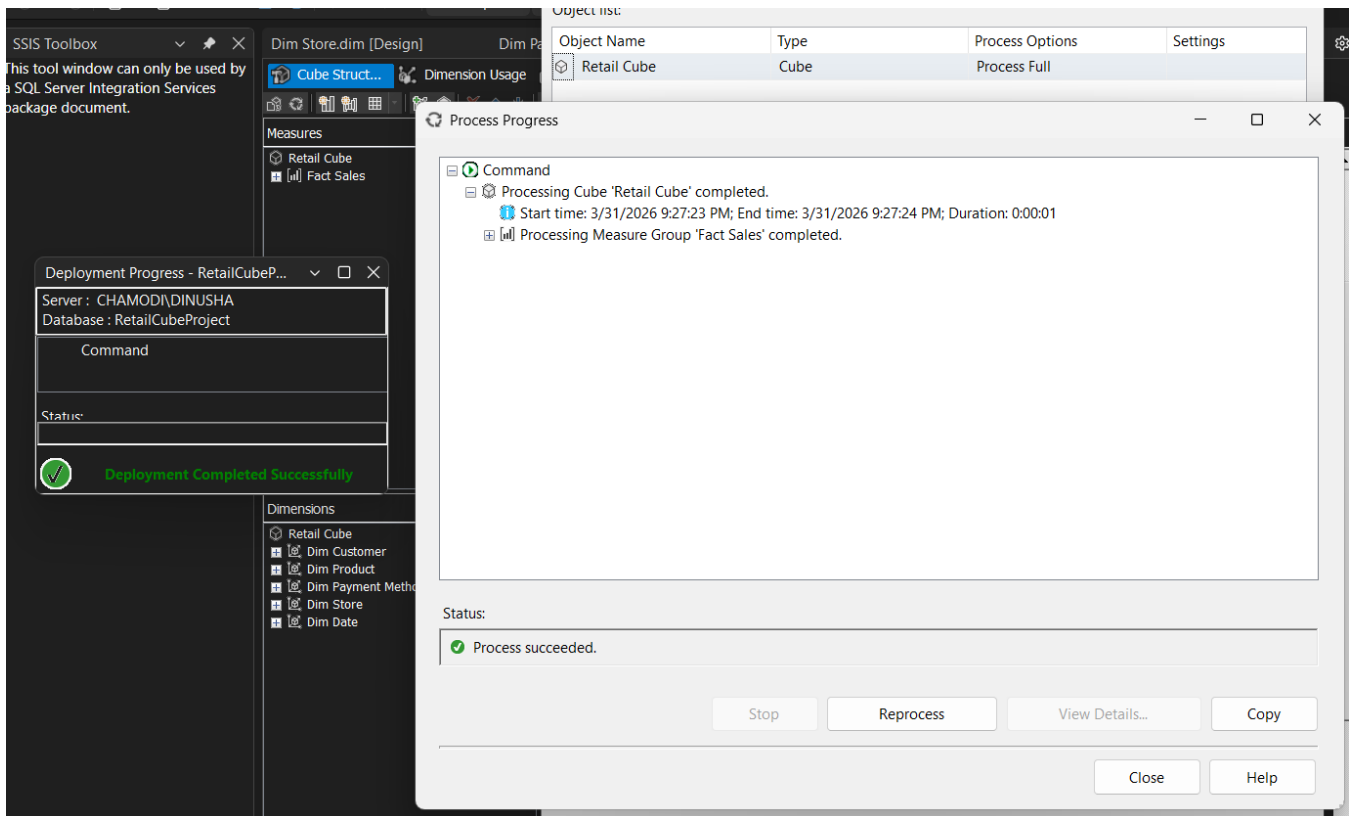
Dimension	Relationship Type	Foreign Key in FactSales
Date (DimDate)	Regular	TransactionDateKey
Customer (DimCustomer)	Regular	CustomerKey
Product (DimProduct)	Regular	ProductKey
Store (DimStore)	Regular	StoreKey
Payment Method (DimPaymentMethod)	Regular	PaymentMethodKey

2.7 Measures in the Fact Table

The following measures were included from the FactSales fact table:

Measure	Aggregation Function	Description
quantity	SUM	Total units sold
price	SUM	Total sales revenue
discount_applied	SUM	Total discount given
txn_process_time_hours	SUM	Transaction processing duration
fac_sales_count	COUNT	Count of rows

2.9 Processing the Cube

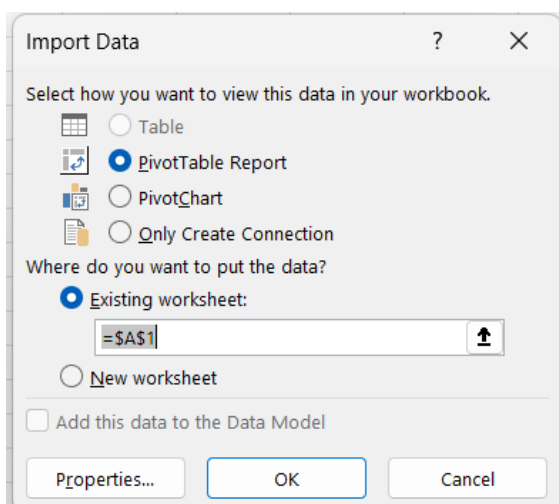
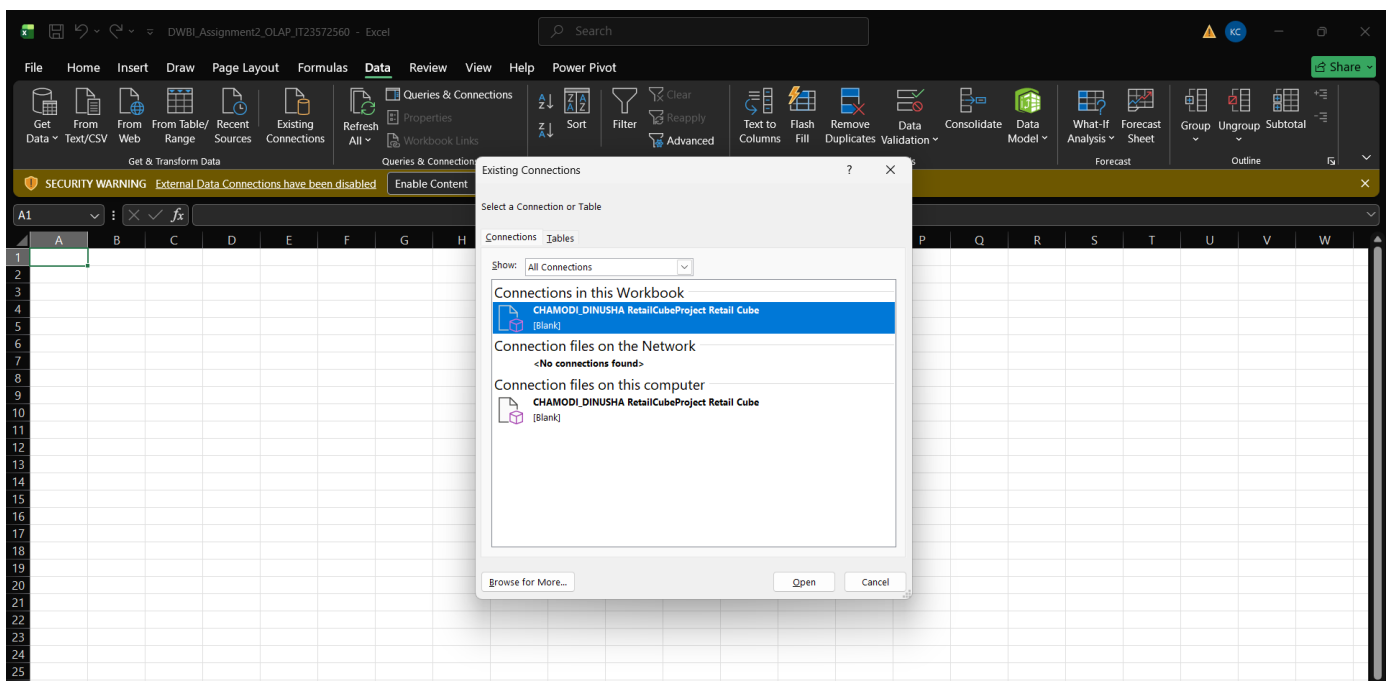


Step 3: Demonstration of OLAP Operations

3.1 Connecting Excel to the SSAS Cube

Microsoft Excel was connected to the SSAS cube to perform OLAP operations using Pivot Tables and Pivot Charts. The connection was established via the Data tab:

1. Open Microsoft Excel and Go to the Data tab on the ribbon to Get and Data From Database and From Analysis Services
2. In the Data Connection Wizard, enter the server name: CHAMODI\NINUSHA and Select the RetailCubeProject database



3.2 OLAP Operations Demonstrated

The following five OLAP operations were demonstrate using Pivot Tables and Pivot Charts connected to the SSAS cube:

3.2.1 Roll-Up

Roll-up aggregates data from lower level of the hierarchy to a higher level, reducing detail and increasing summarization.

Steps performed:

- 1.Placed the Calendar Hierarchy in the Rows area of the Pivot Table
- 2.Added the SUM of Price measure to the Values area

The screenshot displays the Microsoft Excel interface with a PivotTable and the PivotTable Fields task pane. The PivotTable is named 'PivotTable2' and is located in the worksheet 'DWBI_Assignment2_OLAP_IT23572560 - Excel'. The PivotTable has 'Calendar Hierarchy' in the Rows area and 'SUM of Price' in the Values area. The data is summarized by year, with a Grand Total of 19715392.01.

Row Labels	Price
2020	626440.42
2021	2126421.7
2022	3683683.82
2023	5393819.87
2024	6964537.44
2025	920488.76
Grand Total	19715392.01

The PivotTable Fields task pane on the right shows the following fields:

- Fact Sales**
 - ☐ Discount Applied
 - ☐ Fact Sales Count
 - ☒ Price
 - ☐ Quantity
 - ☐ Txn Process Time Hours
- Dim Customer**
 - ☐ City
 - ☐ Customer Key
 - ☐ Full Name

Drag fields between areas below:

- Filters**
- Rows**
 - Calendar Hierarchy

☐ Defer Layout Update

3.2.2 Drill-Down

It navigates from a summarized level to a more detailed level within the hierarchy.

Year → Quarter → Month → Date.

	A	B
1		
2		
3	Row Labels	Price
4	2020	
5	1	4998.52
6	2	53960.47
7	3	178498.17
8	4	388983.26
9	2021	
10	1	315203.66
11	2	347124.07
12	3	572927.03
13	4	891166.94
14	2022	
15	1	545975.21
16	2	705401.21
17	3	1050934.8
18	4	1381372.6
19	2023	
20	1	905973.27
21	2	1099939.8
22	3	1487244
23	4	1900662.8
24	2024	
25	1	1282890.9
26	2	1302821.5
27	3	1916415.9
28	4	2462409.2
29	2025	
30	1	920488.76
31	Grand Total	19715392
32		

B29							19715392.0099998
	A	B	C	D	E	F	
3	Row Labels	Price					
4	2020						
5	1						
6	March						
7	2020-03-04	91.16					
8	2020-03-12	857.83					
9	2020-03-16	651.75					
10	2020-03-23	556.8					
11	2020-03-24	342.82					
12	2020-03-26	438.66					
13	2020-03-27	523.96					
14	2020-03-29	325.67					
15	2020-03-30	223.71					
16	2020-03-31	986.16					
17	2	53960.47					
18	3	178498.17					
19	4	388983.26					
20	2021						
21	1	315203.66					
22	2	347124.07					
23	3	572927.03					
24	4	891166.94					
25	2022	3683683.82					
26	2023	5393819.87					
27	2024	6964537.44					
28	2025	920488.76					
29	Grand Total	19715392.01					

3.2.3 Slice

This select single dimension member to filter the dataset along one dimension, creating a subset view.

Steps performed

1. dragged the Payment Method dimension to the Filters area
2. Selected a specific payment method from the filter dropdown

	A	B
1	Payment Method	Cash 
2		
3	Row Labels 	Price
4	+ 2020	35617.49
5	+ 2021	114764.39
6	+ 2022	185753.35
7	+ 2023	309509.92
8	+ 2024	295950.76
9	+ 2025	44486.39
10	Grand Total	986082.3
11		

	A	B
1	Payment Method	Apple Pay 
2		
3	Row Labels 	Price
4	+ 2020	51343
5	+ 2021	209644.31
6	+ 2022	341117.61
7	+ 2023	539507.87
8	+ 2024	682495.05
9	+ 2025	83374.24
10	Grand Total	1907482.08
11		

	A	B
1	Payment Method	PayPal 
2		
3	Row Labels 	Price
4	+ 2020	92237.33
5	+ 2021	291652.75
6	+ 2022	553727.65
7	+ 2023	806869.72
8	+ 2024	1029695.55
9	+ 2025	145178.1
10	Grand Total	2919361.1
11		

	A	B
1	Payment Method	Google Pay 
2		
3	Row Labels 	Price
4	+ 2020	35346.39
5	+ 2021	106611.83
6	+ 2022	190654.12
7	+ 2023	257538.12
8	+ 2024	346436.46
9	+ 2025	42739.27
10	Grand Total	979326.19
11		

3.2.4 Dice

Dice select sub-cube by applying filters on multiple dimensions simultaneously.

Steps performed

1. Applied multiple filters simultaneously

ex-

◦Product dimension: Smart Phones

◦Date dimension:2022 to 2024

◦Payment Method: Pay pal

	A	B	
1	Product Hierarchy	Smartphones	▼
2	Payment Method	PayPal	▼
3	Calendar Hierarchy	(Multiple Items)	▼
4			
5	Price		
6	431948.49		
7			

ex-

◦Product dimension: TVs

◦Date dimension: 2025.01.04

◦Payment Method:Debit Card

	A	B	
1	Product Hierarchy	TVs	▼
2	Payment Method	Debit Card	▼
3	Calendar Hierarchy	2025-01-04	▼
4			
5	Price		
6	1396.65		
7			

3.2.5 Pivot

Pivot rotates the data view by rearranging the rows and columns, changing the analysis perspective without altering the underlying data.

Steps performed

- Initially placed Product Category in Rows and Calendar Year in Columns, with SUM of Price in Values

The screenshot shows an Excel spreadsheet with a PivotTable. The PivotTable is located in the range B4:H21. The PivotTable Fields task pane is open on the right side of the screen. The task pane shows the following configuration:

- Fact Sales** (checked):
 - ☐ Discount Applied
 - ☐ Fact Sales Count
 - ☒ Price
 - ☐ Quantity
 - ☐ Txn Process Time Hours
- Filters**: (Empty)
- Columns**: Calendar Hierarchy
- Rows**: Product Hierarchy
- Values**: Price
- ☐ Defer Layout Update
- Update** button

The PivotTable data is as follows:

Row Labels	2020	2021	2022	2023	2024	2025	Grand Total
Audio Equipment	16842.15	50374.88	101483.85	143545.1	176271.55	22390.75	510908.28
Bedding	1636.25	9749.56	22778.97	31273.41	32263.52	3845.86	101547.57
Computer Accessories	1543.7	4591.18	9035.89	15272.71	19144.53	1999.45	51587.46
Cookware	4171.2	9680.79	26671.78	38732.25	56960.74	7244.04	143460.8
Desktop Computers	31079.74	63375.31	149186.71	237146.62	293769.2	62789.83	837347.41
Furniture	120663.41	468002.38	807784.17	1162390.44	1520317.42	169710.89	4248868.71
Gaming Consoles	22070.15	76385.13	139770.33	181276.67	266303.69	35610.17	721416.14
Home Decor	17033.91	58176.33	116890.27	161040.59	202288.55	28850.24	584279.89
Kitchen Appliances	61836.41	210980.17	392611.8	530119.81	731186.1	85865.61	2012599.9
Laptops	102046.07	299638.67	500776.79	820638.15	988610.37	114521.87	2826231.92
Small Kitchen Appliances	4516.13	15549.7	26873.2	39178.15	53689.53	7755.51	147562.22
Smart Home Devices	20968.19	77898.09	130518.22	189540.23	254590.66	34831.9	708347.29
Smartphones	105020.83	381642.48	636541.95	921962.19	1232880.26	190739.9	3468787.61
Tablets	7034.58	39442.47	60876.8	92564.79	109726.92	16337.54	325983.1
TVs	77787.95	267924.05	404241.86	620622.54	753444.41	92056.11	2216076.92
Unknown	32189.75	93010.51	157641.23	208516.22	273089.99	45939.09	810386.79
Grand Total	626440.42	2126421.7	3683683.82	5393819.87	6964537.44	920488.76	19715392.01

The screenshot shows an Excel spreadsheet with a PivotTable. The PivotTable is located in the range B6:P24. The PivotTable Fields task pane is open on the right side of the screen. The task pane shows the following configuration:

- Fact Sales** (checked):
 - ☐ Discount Applied
 - ☐ Fact Sales Count
 - ☒ Price
 - ☐ Quantity
 - ☐ Txn Process Time Hours
- Filters**: (Empty)
- Columns**: Calendar
- Rows**: Product
- Values**: Price
- ☐ Defer Layout Update
- Update** button

The PivotTable data is as follows:

Row Labels	2020-03-04	2020-03-12	2020-03-16	2020-03-23	2020-03-24	2020-03-26	2020-03-27	2020-03-29	2020-03-30	2020-03-31
Audio Equipment	91.16	159.8							1360.94	4412.22
Bedding									596.53	309.89
Computer Accessories									140.47	436.04
Cookware									141.55	1129.98
Desktop Computers									515.58	14382.21
Furniture									6313.64	39959.62
Gaming Consoles									2295.02	5617.58
Home Decor									737.97	6229.04
Kitchen Appliances									986.16	4986.36
Laptops									14411.1	23700.31
Small Kitchen Appliances									828.13	1384.7
Smart Home Devices									1898.36	4774.49
Smartphones									10825.02	29346.95
Tablets									336.07	3197.88
TVs									6418.23	16837.04
Unknown									2155.5	8233.77
Grand Total	91.16	857.83	651.75	556.8	342.82	438.66	523.96	325.67	986.16	53960.47

- Dragged Calendar Year from Columns to Rows, Dragged Product Category from Rows to Columns

Step 4: Power BI Reports

4.1 Connecting Power BI Desktop to RetailDW

Power BI Desktop was connected to the RetailDW SQL Server data warehouse to import the required tables for reporting:

1. Open Power BI Desktop and connect

- Server: CHAMODI
- Database: RetailDW

2. Select Import as the data connectivity mode and Authenticate using Windows credentials: chamodi\msi

3. In the Navigator, select the following tables:

- FactSales
- DimDate

- DimCustomer
- DimProduct
- DimStore
- DimPaymentMethod

5.Click Load to import all selected tables into Power BI

4.2 Data Modeling in Power BI

After loading the tables, the relationships were verified in the Model View:

Dimension Table	Fact Table	Join Key
DimDate	FactSales	DateKey → TransactionDateKey
DimCustomer	FactSales	CustomerKey
DimProduct	FactSales	ProductKey
DimStore	FactSales	StoreKey
DimPaymentMethod	FactSales	PaymentMethodKey

4.3 Report 1: Matrix Visual Report

Create a report with a Matrix visual to display detailed tabular data with row and column groupings.

4.3.1.Steps Followed

- 1.In Power BI Desktop, created a new report page and named it Report 1 – Matrix Visual
- 2.From the Visualizations pane, selected the Matrix visual
- 3.Configured the Matrix fields as follows:
 - Rows: product_category (from DimProduct) → product_name (creating a row hierarchy)
 - Columns: CalendarYear (from DimDate)

◦Values: SUM of price (from FactSales)

product_name	2020	2021	2022	2023	2024	2025	Total
Air Fryer	355.05	2,370.79	7,112.11	7,330.72	10,180.31	886.29	28,235.27
Amazon Echo	3,889.16	19,941.04	27,358.33	35,089.27	51,691.54	11,335.00	149,304.34
Amazon Fire HD	1,005.56	13,441.44	15,048.07	19,904.31	18,133.74	3,085.22	70,618.34
Area Rug	3,075.26	10,337.45	24,250.10	32,154.88	44,064.02	5,577.16	119,458.87
Asus ROG	4,917.84	11,716.36	38,066.96	47,813.08	73,206.95	6,699.91	182,421.10
Asus ZenBook	25,335.88	67,663.34	88,385.04	145,897.66	207,588.50	20,059.45	554,929.87
Audio-Technica Turntable	4,439.73	14,614.60	23,422.16	27,881.27	38,482.56	5,046.24	113,886.56
Baking Sheet	975.73	1,606.15	5,175.88	8,716.04	12,573.32	1,536.04	30,583.16
Bed Frame	34,255.26	113,020.25	172,950.39	249,438.83	315,842.70	35,662.37	921,169.80
Blender	1,174.52	2,904.46	4,779.91	7,152.29	8,558.28	1,061.83	25,631.29
Bookshelf	29,437.23	80,716.08	158,903.23	230,580.72	270,199.35	35,284.88	805,121.49
Bose Headphones	2,223.32	8,904.07	20,327.45	28,313.65	30,521.77	4,319.40	94,609.66
Cast Iron Skillet	141.55	712.77	5,241.00	7,199.28	11,500.77	279.08	25,074.45
Coffee Maker	965.97	3,889.72	4,122.04	8,301.40	10,952.32	2,691.46	30,922.91
Comforter Set	220.78	1,857.23	4,322.84	6,997.59	4,951.55	1,324.60	19,674.59
Cookware Set	1,075.28	1,941.40	7,091.74	8,614.92	10,160.06	4,081.98	32,965.38
Curtains	3,401.38	12,230.34	22,214.71	31,222.05	38,444.88	4,786.80	112,300.16
Dell Inspiron Desktop	11,743.14	12,612.84	34,270.12	60,455.57	52,396.42	8,639.02	180,117.11
Dell XPS 15	22,101.16	61,986.88	110,095.89	177,144.85	170,605.61	14,304.24	556,238.63
Dining Table	13,138.53	91,089.30	153,630.02	213,435.18	314,023.87	27,513.52	812,830.42
Dishwasher	7,893.01	40,790.49	84,267.44	85,914.52	156,026.19	18,391.14	393,282.79
Dutch Oven	749.66	3,461.17	5,628.50	7,504.20	10,710.44	617.78	28,671.75
Duvet Cover	628.97	2,407.62	6,656.64	4,445.00	8,002.87	695.37	22,836.47
Electric Range	14,070.84	39,898.84	67,766.36	114,605.50	146,475.97	22,548.04	405,365.55
External Hard Drive	159.03	1,036.63	1,709.12	3,127.82	3,542.62	785.64	10,360.86
Food Processor	1,616.77	3,555.07	5,655.89	8,698.91	12,395.92	1,785.55	33,708.11
Google Nest	3,606.13	15,911.01	31,493.04	39,021.87	54,927.46	6,587.24	151,546.75
Google Pixel 6	23,153.96	74,313.49	116,077.39	162,798.25	266,610.22	32,087.19	675,040.50
HP Pavilion	6,646.49	15,232.98	33,310.34	42,491.86	60,649.19	19,251.93	177,582.79
HP Spectre	18,660.38	63,667.99	94,140.77	152,839.91	211,848.56	20,312.30	561,469.91
iMac	3,116.30	14,385.43	18,397.08	41,174.10	56,598.83	12,333.84	146,005.58
iPad Pro	1,402.98	7,164.94	10,130.21	22,418.73	22,741.60	3,659.13	67,517.59
Total	626,440.42	2,126,421.70	3,683,683.82	5,393,819.87	6,964,537.44	920,488.76	19,715,392.01

The Matrix visual display total sales (price) broke down by product category and product name in rows, and by calendar year in columns. This provide comprehensive cross tabular view of how each product performed across each year, with row subtotals per category and a grand total column.

4.4 Report 2: Slicers with Cascading Filters and Multiple Visuals

4.4.1.Data Preparation

To implement cascading filters between the Product Category slicer and the Product Name slicer, the relationship between DimProduct.product_category and DimProduct.product_name was leveraged.since a both fields exist in the same DimProduct table, when the Category slicer fillters the table, the name slicer automaticallly update to show only products within the selected category.

4.4.2. Steps Followed

1.Added Slicer 1 – Product Category:

- Visual: Slicer
- Field: product_category from DimProduct
- Style: List (vertical dropdown)

2.Added Slicer 2 – Product Name:

- Visual: Slicer
- Field: product_name from DimProduct
- Style: List

Visual Type	Fields Used	Insight Shown
Line Chart	CalendarYear (X-axis) vs SUM of price (Y-axis)	Yearly sales trend over 2020–2025
Pie Chart	payment_method (Legend) vs SUM of price (Values)	Sales distribution by payment method
Stacked Column Chart	CalendarYear (X-axis), product_category (Legend), SUM of price (Y-axis)	Category-wise sales by year
Card (KPI)	SUM of price	Total Sales KPI – single number summary

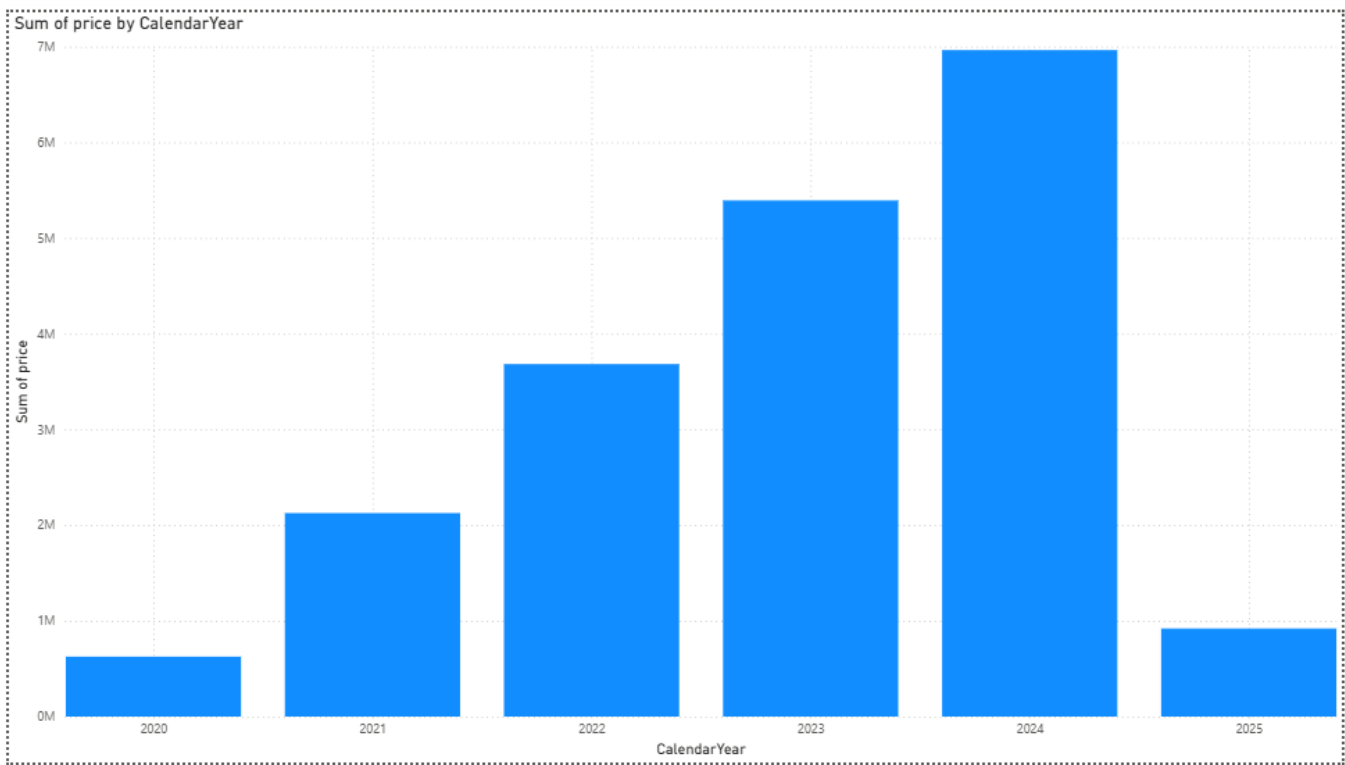
4.5 Report 3: Drill-Down Hierarchical Report

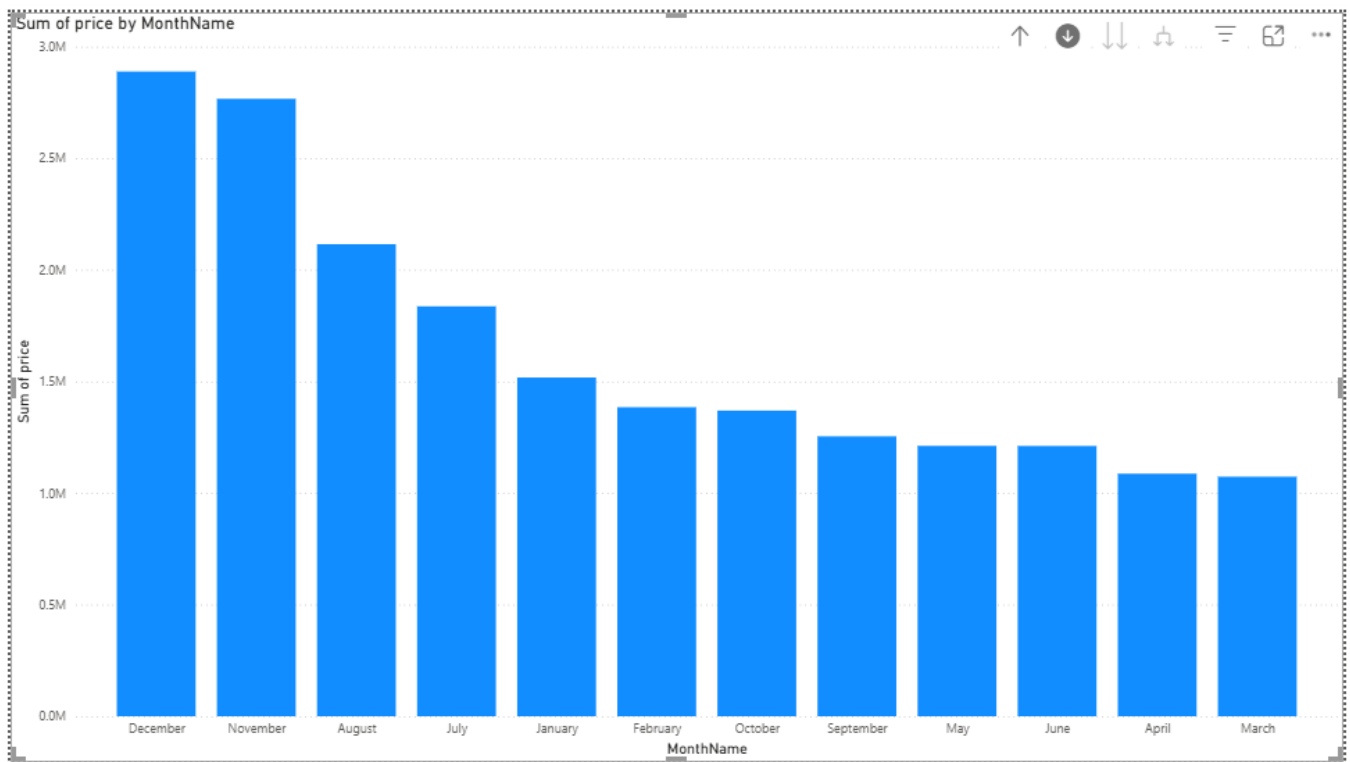
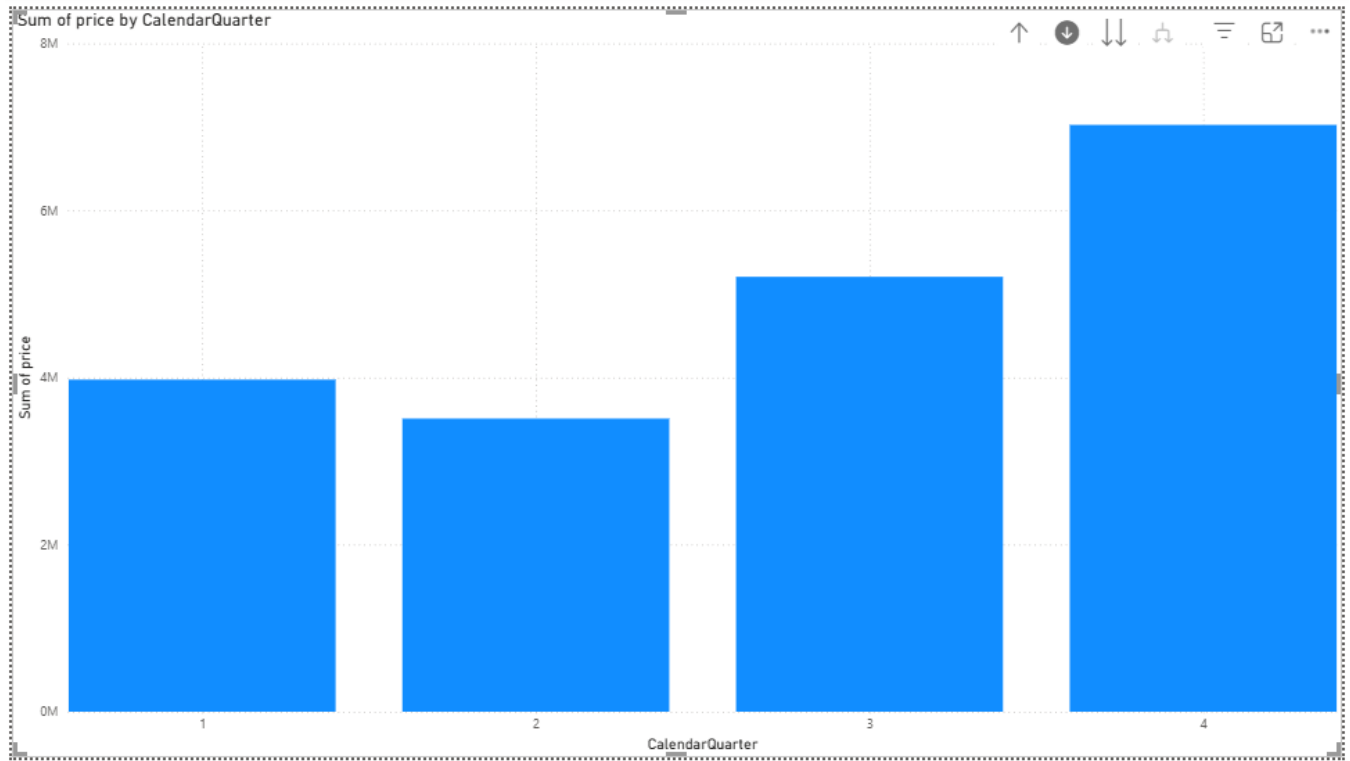
4.5.1 Steps Followed

- Created report page and add clustered column chart with date hierarchy in x axis using year, quarter and month, and set y axis as sum of price
- Enabled drill down and also add line chart with same hierarchy and sum of quantity for compare sales and volume.

4.5.2 Navigation

- Use drill down arrow for enable drill mode, then click year for go to quarter and quarter go to month level.
- Use drill up button for go back and also use next level option for drill all data together



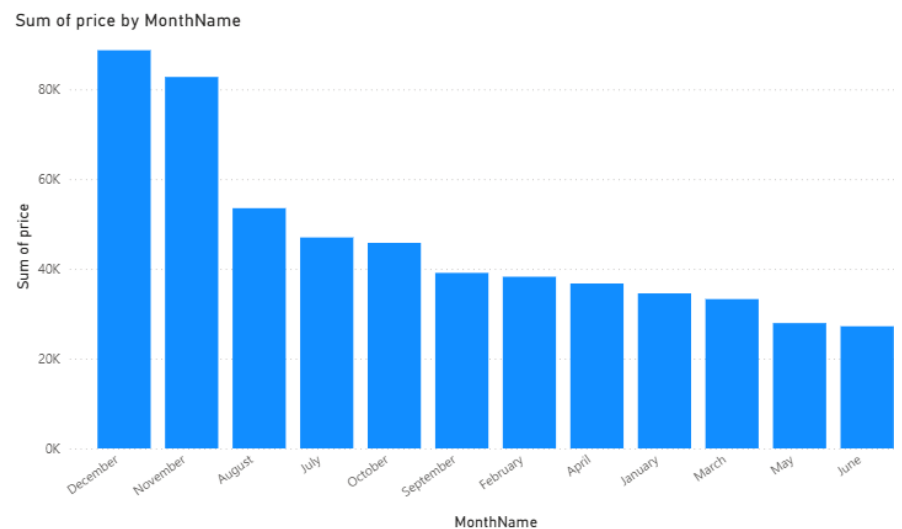


4.6 Report 4: Drill-Through Report

4.6.1 Steps Followed

- Created summary page with bar chart showing sum of price by product category and created drill through detail page
- Set drill through using product category and added table, card and monthly chart for show detailed view
- Drill through done by right click on category and navigate to detail page with filtered data

product_name	Sum of price	Sum of quantity
Asus ZenBook	554,929.87	523
Total	554,929.87	523



554.93K

Sum of price

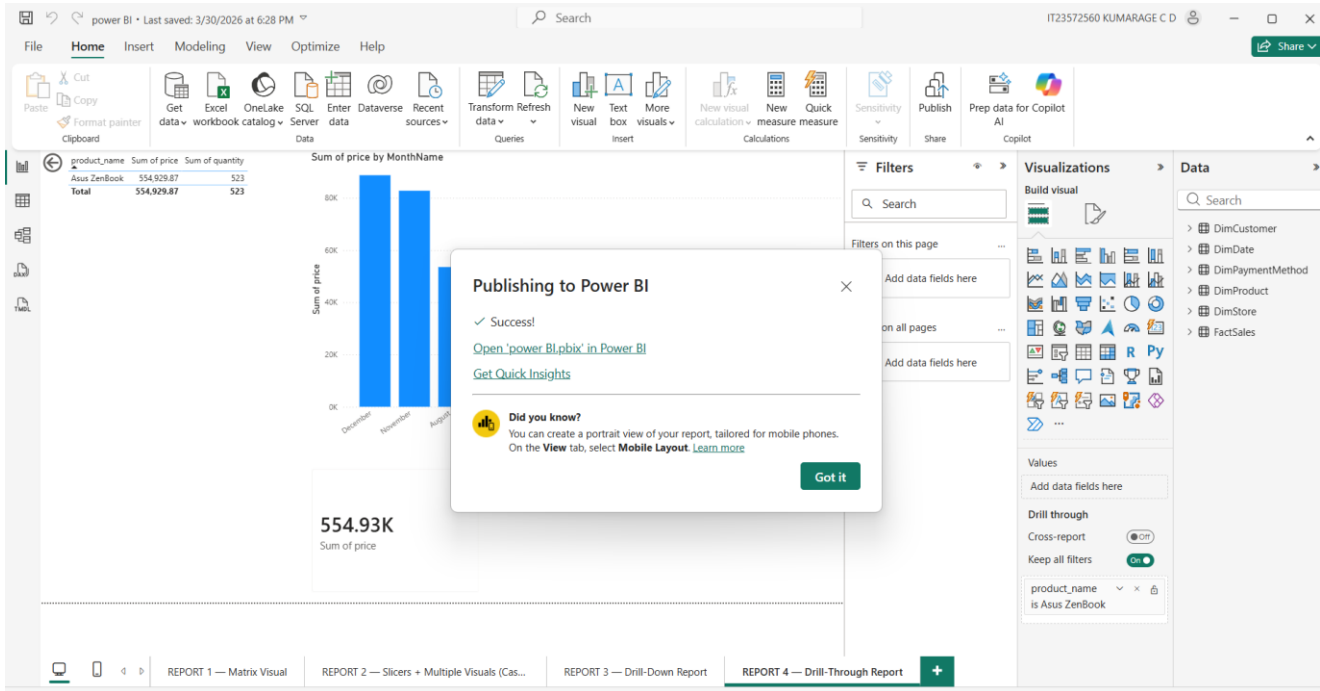
4.6.2 DAX Measures

No additional DAX measures were explicitly required since standard aggregations (SUM) were sufficient. All calculations are handled using Power BI's implicit measures.

4.8 Publishing to Power BI Service

4.8.1.Steps Followed

- Click publish in power bi desktop and sign in, then select workspace and upload report
- Open power bi service and check all reports working correct and showing proper data online.



The screenshot shows the Power BI Service web portal. A matrix report titled 'REPORT 1 — Matrix Visual' is displayed, showing sales data by product and year. The data is organized into a table with columns for product_name, year (2020-2025), and Total. The total sales for 2025 are 19,715,392.01.

product_name	2020	2021	2022	2023	2024	2025	Total
Air Fryer	355.05	2,370.79	7,112.11	7,330.72	10,180.31	886.29	28,235.27
Amazon Echo	3,889.16	19,941.04	27,358.33	35,089.27	51,691.54	11,335.00	149,304.34
Amazon Fire HD	1,005.56	13,441.44	15,048.07	19,904.31	18,133.74	3,085.22	70,618.34
Area Rug	3,075.26	10,337.45	24,250.10	32,154.88	44,064.02	5,577.16	119,458.87
Asus ROG	4,917.84	11,716.36	38,066.96	47,813.08	73,206.95	6,699.91	182,421.10
Asus ZenBook	25,335.88	67,663.34	88,385.04	145,897.66	207,588.50	20,059.45	554,929.87
Audio-Technica	4,439.73	14,614.60	23,422.16	27,881.27	38,482.56	5,046.24	113,886.56
Baking Sheet	975.73	1,606.15	5,175.88	8,716.04	12,573.32	1,536.04	30,583.16
Bed Frame	34,255.26	113,020.25	172,950.39	249,438.83	315,842.70	35,662.37	921,169.80
Blender	1,174.52	2,904.46	4,779.91	7,152.29	8,558.28	1,061.83	25,631.29
Bookshelf	29,437.23	80,716.08	158,903.23	230,580.72	270,199.35	35,284.88	805,121.49
Bose Headphones	2,223.32	8,904.07	20,327.45	28,313.65	30,521.77	4,319.40	94,609.66
Cast Iron Skillet	141.55	712.77	5,241.00	7,199.28	11,500.77	279.08	25,074.45
Coffee Maker	965.97	3,889.72	4,122.04	8,301.40	10,952.32	2,691.46	30,922.91
Comforter Set	220.78	1,857.23	4,322.84	6,997.59	4,951.55	1,324.60	19,674.59
Cookware Set	1,075.28	1,941.40	7,091.74	8,614.92	10,160.06	4,081.98	32,965.38
Curtains	3,401.38	12,230.34	22,214.71	31,222.05	38,444.88	4,786.80	112,300.16
Dell Inspiron Desktop	11,743.14	12,612.84	34,270.12	60,455.57	52,396.42	8,639.02	180,117.11
Dell XPS 15	22,101.16	61,986.88	110,095.89	177,144.85	170,605.61	14,304.24	556,238.63
Dining Table	13,138.53	91,089.30	153,630.02	213,435.18	314,023.87	27,513.52	812,830.42
Dishwasher	7,893.01	40,790.49	84,267.44	85,914.52	156,026.19	18,391.14	393,282.79
Dutch Oven	749.66	3,461.17	5,628.50	7,504.20	10,710.44	617.78	28,671.75
Duvet Cover	628.97	2,407.62	6,656.64	4,445.00	8,002.87	695.37	22,836.47
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Total	626,440.42	2,126,421.70	3,683,683.82	5,393,819.87	6,964,537.44	920,488.76	19,715,392.01

